Dr. Riggins **AE 5535** **Spring 2024**
184 Toomey **Class: MWF 11:00 – 11:50 am 256 Toomey**
riggins@mst.edu **Office hours: 12:30 – 2:00 pm, Mon, Wed**

Aerospace Propulsion Systems

THIS IS A NOTE-INTENSIVE COURSE! MUCH OF THE MATERIAL IS (DIRECTLY) GIVEN ONLY IN THE COURSE NOTES! (I.E. VIEW THE TEXTBOOK AS 'SUPPLEMENTARY' TO THE NOTES – BUT STILL HIGHLY RECOMMENDED).

I will try to upload notes in 'segments' to CANVAS (under 'file' tab), prior to my going through them in class, but if you do not attend class, I am not responsible for you not understanding the notes – that is on you. So …

YOU ARE RESPONSIBLE FOR ATTENDING CLASS, UNDERSTANDING THE NOTES THAT I GO OVER AND ASKING QUESTIONS DURING CLASS, AND TAKING ANY ADDITIONAL NOTES ON THE MATERIAL AS I GO THROUGH THE UPLOADED NOTES RELEVANT TO THAT MATERIAL.

Text: Mainly notes, but Aerothermodynamics of Gas Turbines and Rocket Propulsion: Gordon Oates **(Good reference but not required)**; mainly strong in jet engine analysis, light on rockets

Other good texts: Mechanics and Thermodynamics of Propulsion, Hill and Peterson (mainly rockets/jet engine components/ no jet engine off-design material covered)
(mainly jet engines, light on rockets)
Elements of Propulsion: Gas Turbines and Rockets; Mattingly (mainly strong in jet engine analysis, light on rockets)
Rocket Propulsion Elements: Sutton (only rockets)

Course Topics:

Control Volume Problems and Fluids/Propulsion Review	Notes, hand-outs, AE 4435 work
Non-Ideal Engines	Notes, Chapter 7
Engine Off-Design Performance and Component Matching	Notes; Chapter 8
Turbo-Machinery Aerodynamics	Notes; Chapter 9
Chemical Rockets	Chapter 3; handouts, notes
Non-Chemical Rockets and Space Missions	Chapter 3; handouts, notes
Turboprop Analysis	Chapter 7; notes
Power Generation in Space	handouts, notes

Ramjet/Scramjet Engine Design (time permitting)	handouts, notes
Innovative Hybrid & Combined-Cycle (Rocket-Airbreather) Aerospace Engines (time permitting)	handouts, notes
Rocket Nozzles/Aero-Spike and Plug Nozzle Analysis	handouts, notes
Introduction to Computational Fluid Dynamics (time permitting)	handouts, notes

Homework	10%
Projects (3)	18% (each)
Test	18%
Final Exam	18%

COURSE PROCEDURE

As noted, CANVAS will be used to post instructor notes, prior to material being covered in class. You are responsible for downloading these notes (if you wish). Other than uploading notes, I DO NOT PLAN ON USING CANVAS. Assignments will be provided to you, collected when due, and handed back out when graded in the classroom, during class time. Grade computations will be done only at the end of the semester, and not on CANVAS. You should archive all completed and graded work yourself.

Assignments: LATE WORK OR EMAILED WORK WILL NOT BE ACCEPTED UNLESS AUTHORIZED.

When preparing work, make sure the writing is legible and neat, with clear formulation and good organization. Follow all directions and provide all requested information and background work, especially for high-value project, test, and final exam.

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Contact: equity@mst.edu (Equity and Title IX Office for Missouri S&T)